Government College of Engineering, Jalgaon (An Autonomous Institute of Government of Maharashtra)	
Name :	**PRN :**
Class : L. Y. B.Tech Computer	**Batch:**
Subject : CO456U Cloud Computing Lab	**Sem :** VIIIth
Course Teacher: Prof. S. D. Cheke	**Academic Year :** 2025-26
Date of Performance :	**Date of Completion :**

Practical No - 5

Aim:- To create an AWS S3 bucket and deploy a static one-page website on it, and demonstrate uploading files to the S3 bucket using a Python script.

Theory:-

Amazon S3 (Simple Storage Service) is an object storage service provided by Amazon Web Services that allows users to store and retrieve data from anywhere on the internet.

S3 supports static website hosting, which allows users to host HTML, CSS, and JavaScript files without using a traditional web server.

Benefits of S3 Static Hosting:

- Highly scalable
- Low cost
- Easy deployment
- Integrated with other AWS services

Python scripts can interact with AWS services using the **AWS SDK for Python (Boto3)** provided by Amazon.

Requirements:-

Software

- AWS Account
- Python 3.x
- AWS CLI installed
- Internet browser

Python Libraries

- boto3

Install using:

```
pip install boto3
```

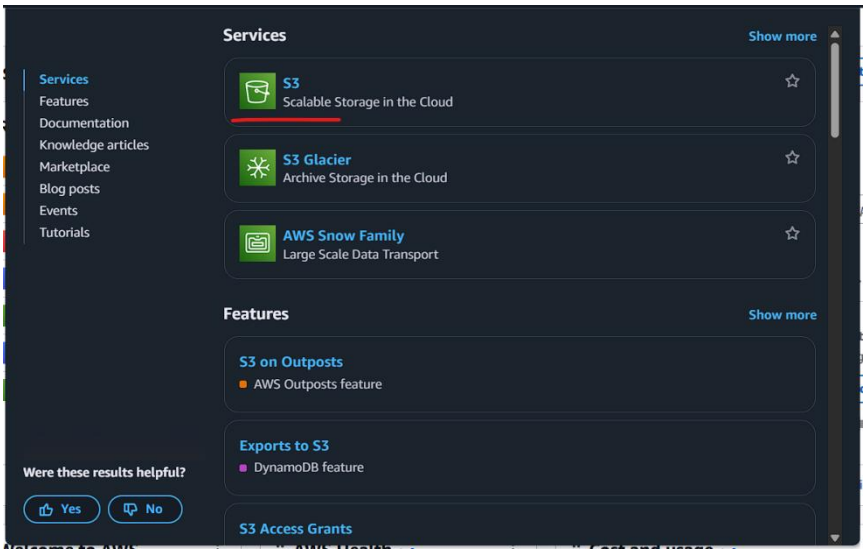
Architecture Diagram:-

User Browser
|
v
AWS S3 Bucket (Static Website Hosting Enabled)
|
v
HTML / CSS / JS Files
|
v
Uploaded using Python Script (Boto3)

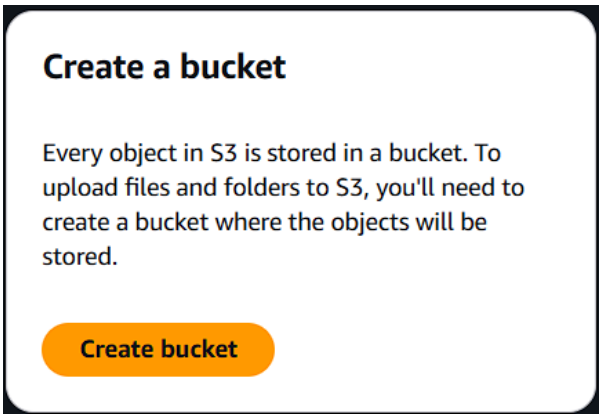
Procedure:-

Part 1: Create an S3 Bucket

1. Login to the AWS Management Console.
2. Open the S3 Service in Amazon Web Services.

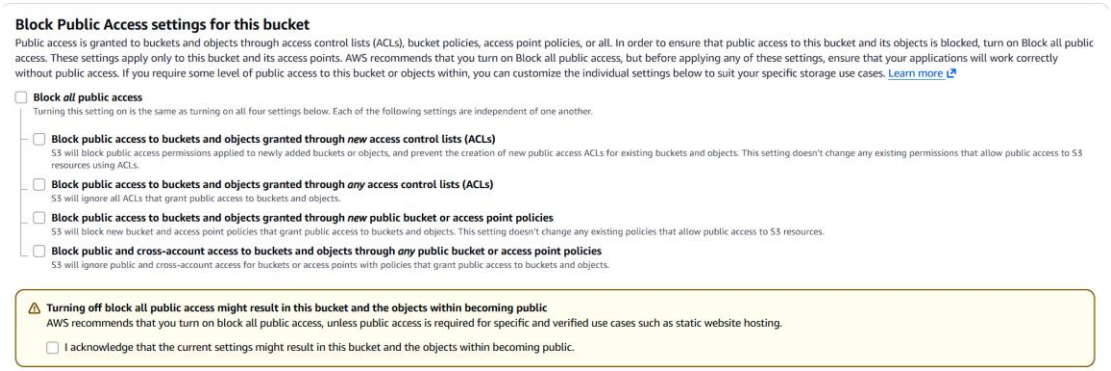


3. Click Create Bucket.



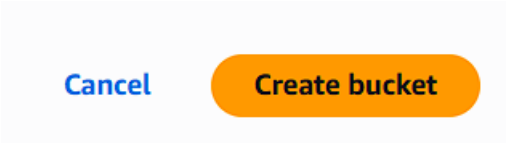
4. Enter:
    1. Bucket Name: unique name
       Example: my-static-website-bucket123

5. Disable "Block Public Access" (required for public website hosting).



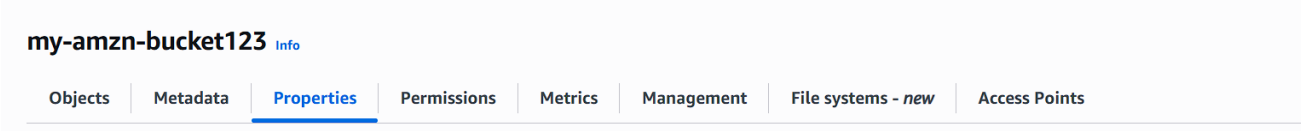
6. Acknowledge the warning.

7. Click Create Bucket.

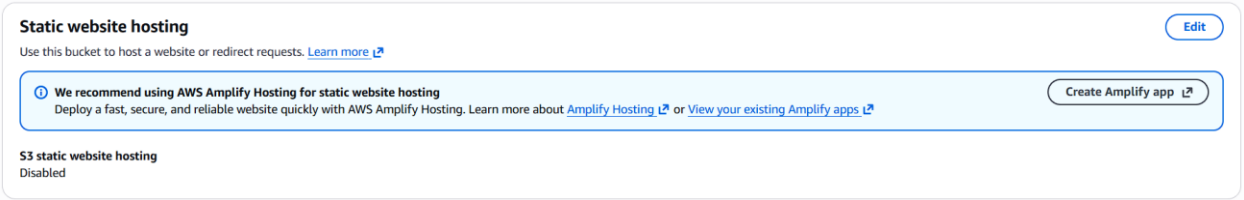


Part 2: Enable Static Website Hosting

1. Open the created bucket.
2. Go to **Properties**.

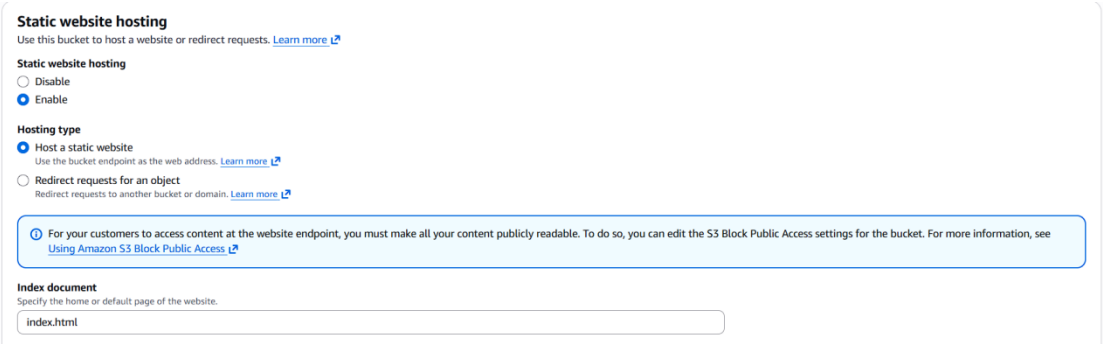


3. Scroll to **Static Website Hosting**.



4. Click **Edit**.
5. Select: Enable

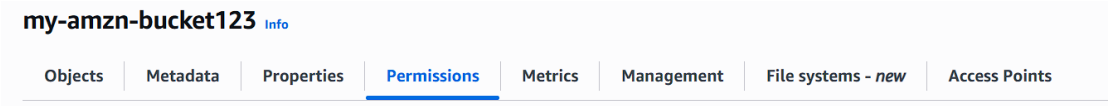
In Index document type index.html



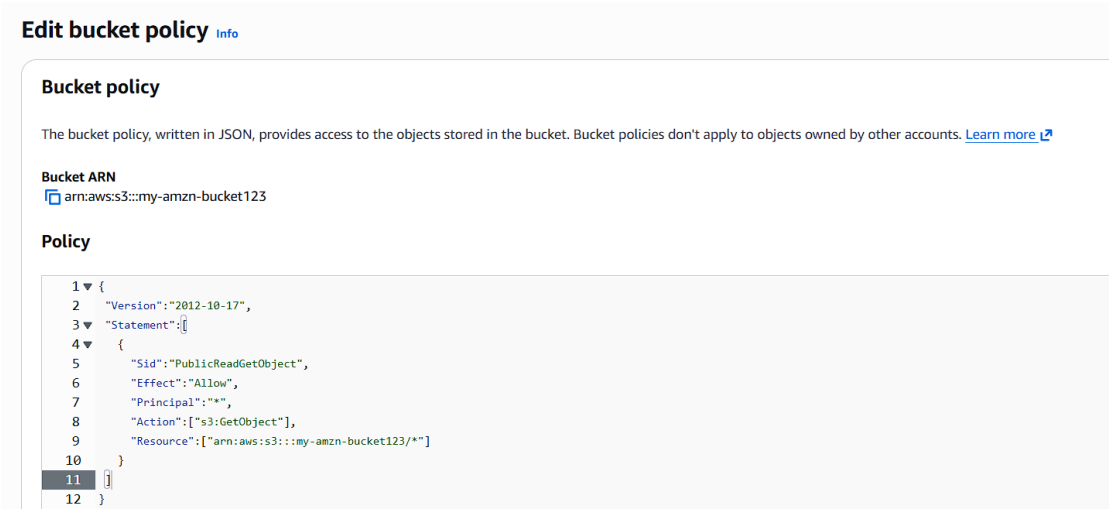
Save Changes

Part 3: Set Bucket Policy for Public Access

Go to **Permissions → Bucket Policy** and paste:




```
{
 "Version":"2012-10-17",
 "Statement":[
   {
     "Sid":"PublicReadGetObject",
     "Effect":"Allow",
     "Principal":"*",
     "Action":["s3:GetObject"],
     "Resource":["arn:aws:s3:::my-static-website-bucket123/*"]
   }
 ]
}
```



Replace the bucket name accordingly in “Resource”.

Save the policy.

Part 4: Create Static Web Page

Create a file **index.html**

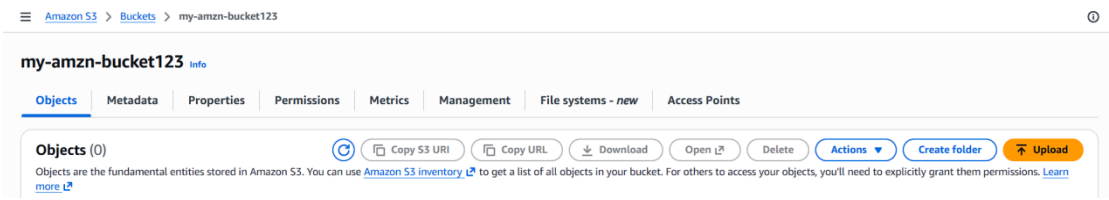
```
<!DOCTYPE html>
<html>
<head>
<title>My First S3 Website</title>
</head>

<body>

<h1>Welcome to My Static Website</h1>
<p>This website is hosted on AWS S3.</p>

</body>
</html>
```

index.html > html

```html
<!DOCTYPE html>
<html>
<head>
<title>My First S3 Website</title>
</head>

<body>

<h1>Welcome to My Static Website</h1>
<p>This website is hosted on AWS S3.</p>

</body>
</html>
```

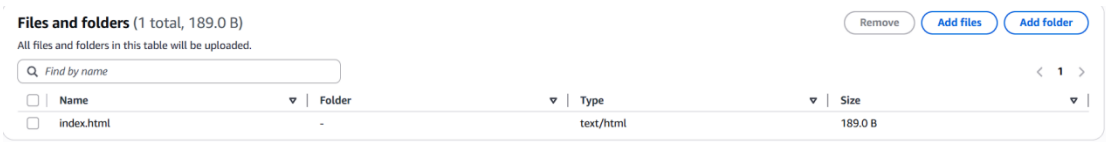
Part 5: Upload Files Manually

1. Open S3 Bucket.
2. Click **Upload**.

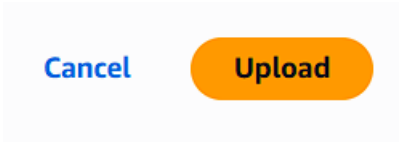


3. Click on “Add files” or “Add folder”

4. Select index.html from your pc



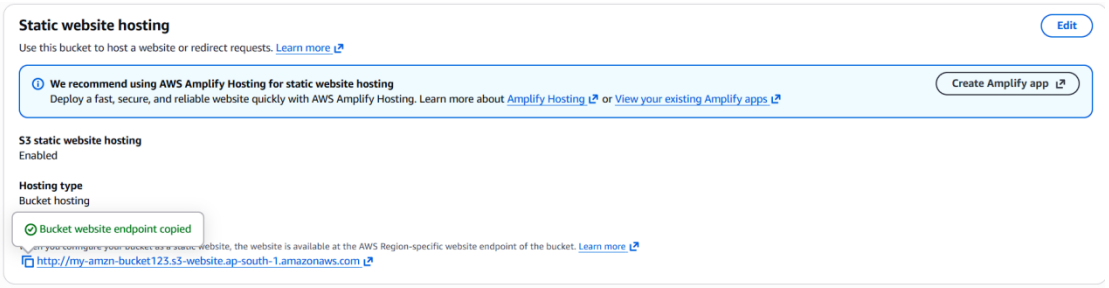
5. Click Upload



Part 6: Access the Website

Go to:

Properties → Static Website Hosting



Copy the **Endpoint URL**.

Example:

`http://my-static-website-bucket123.s3-website-ap-south-1.amazonaws.com`

Open it in the browser.

Your **static website will load**.

Welcome to My Static Website

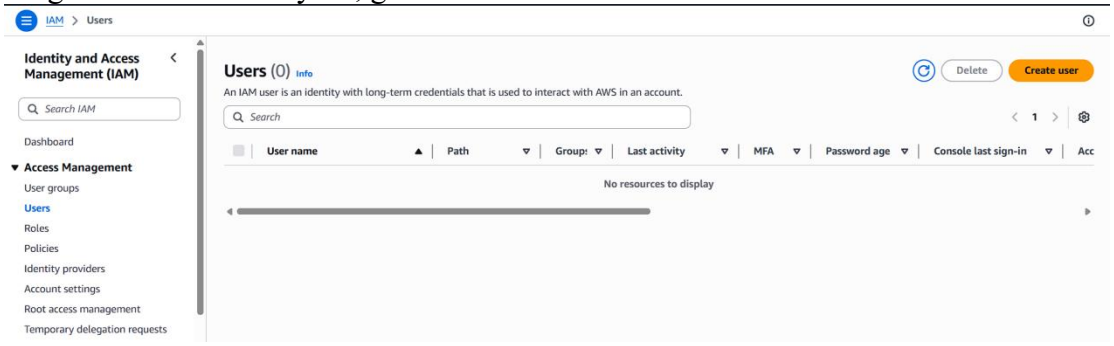
This website is hosted on AWS S3.

Part 7: Upload Files Using Python Script

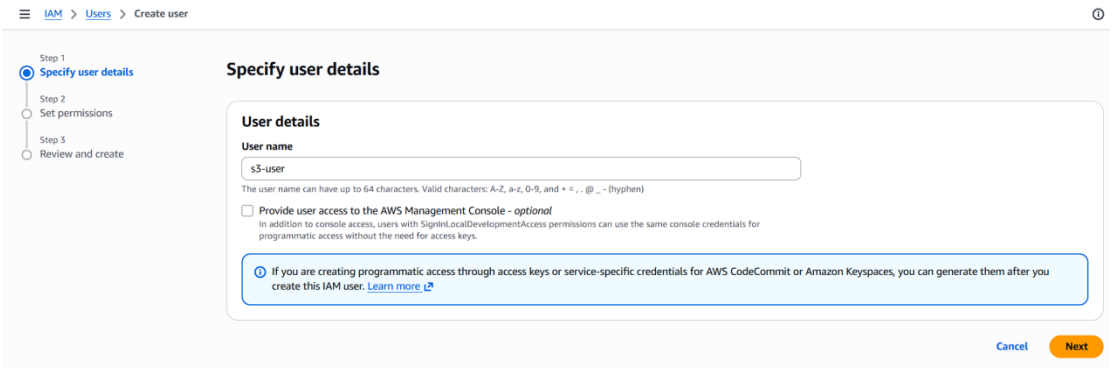
Step 1: Install awscli from
"https://docs.aws.amazon.com/cli/latest/userguide/getting-started-install.html"

1. Use aws --version to check the version in cmd prompt
2. Type aws configure in cmd prompt

To get AWS Acces Key ID, go to aws console and search IAM and click on it.



Step 2 : Go to left panel and select Users. Then click "Create User" and type a user name of your choice.

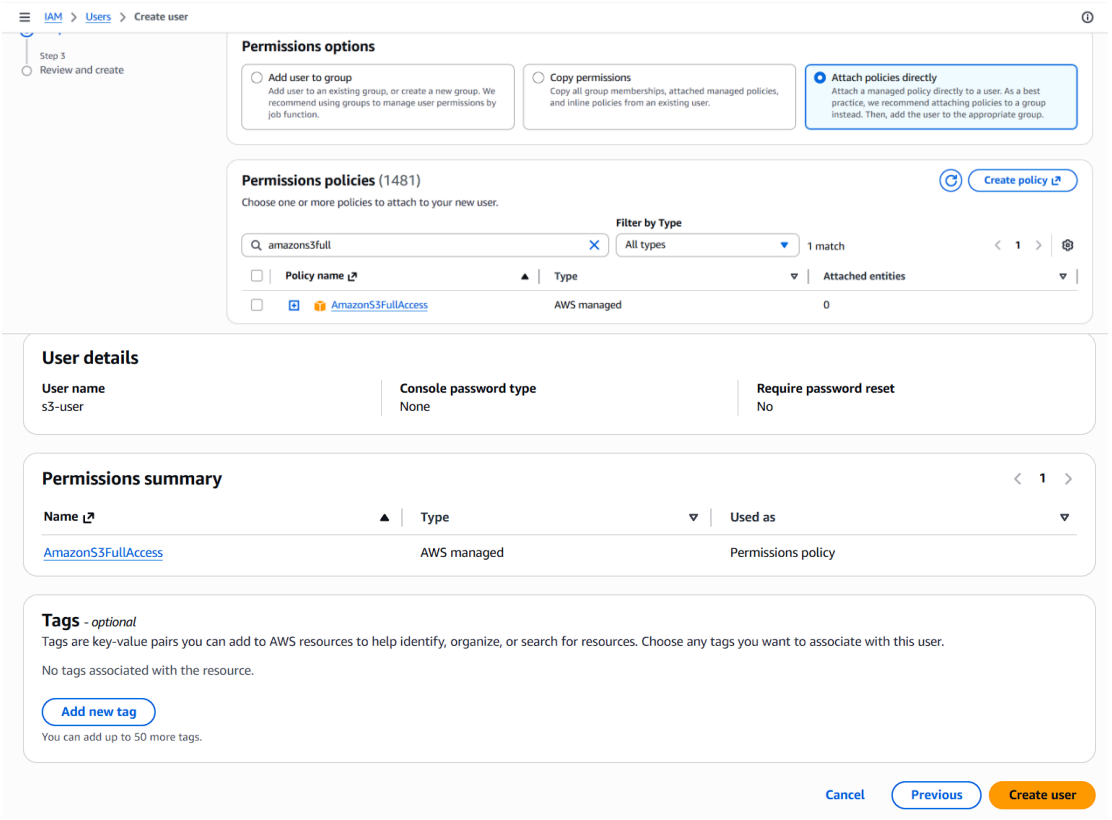


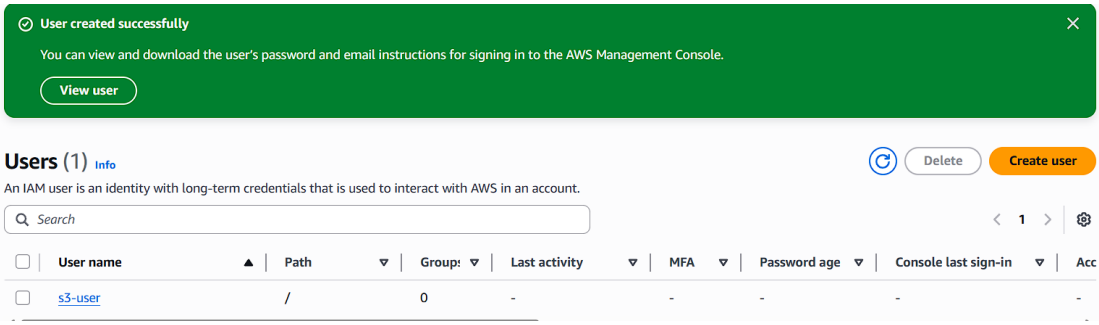
Click “Next”.

Step 3: Give Permissions

1. Select **Attach policies directly**.
2. Search and select: AmazonS3FullAccess

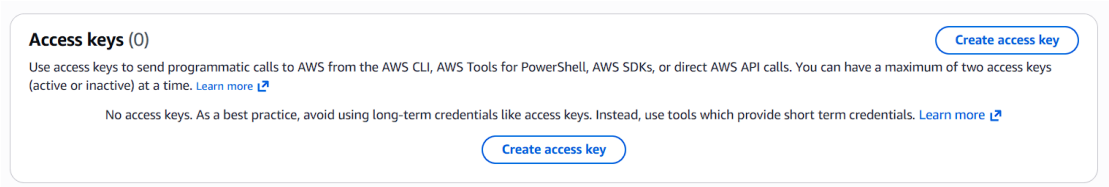
Click Next → Create user.





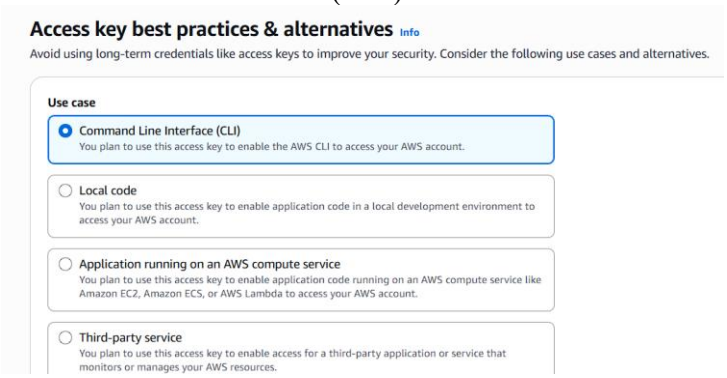
Step 4: Create Access Key

1. Click the user you created.
2. Go to Security Credentials tab.
3. Scroll to Access Keys.
4. Click Create Access Key.



Choose:

Command Line Interface (CLI)



Click Next → Create Access Key.

Now copy the Access key and Secret Access key in the terminal where it is required.
Default region name: (shown in the top right corner of your aws console)
Default output format: json

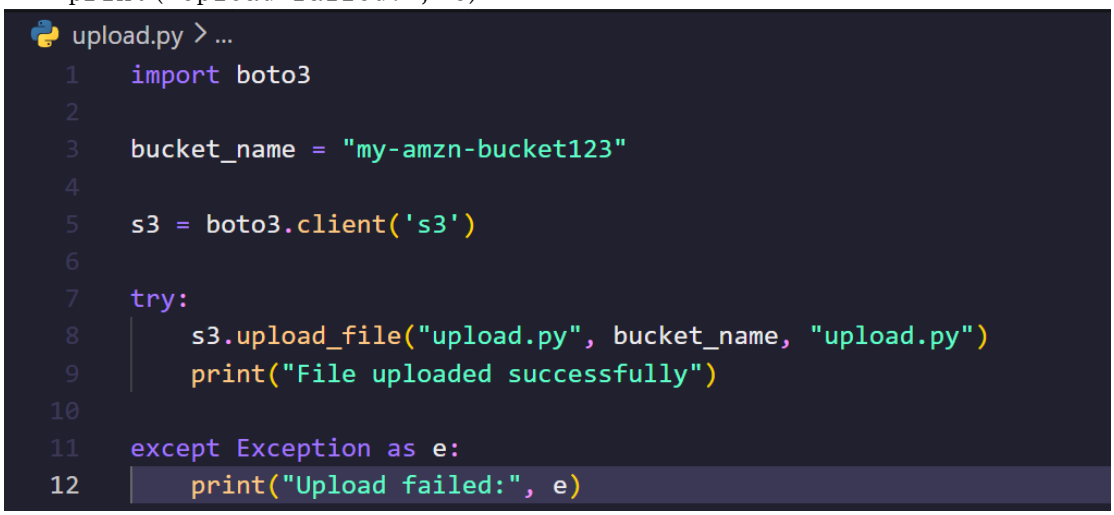
Step 2: Python Script to Upload File

Create file:

upload.py

Python code:

```
import boto3

bucket_name = "my-static-website-bucket123"

s3 = boto3.client('s3')

try:
    s3.upload_file(file_name, bucket_name, "file_name")
    print("File uploaded successfully")

except Exception as e:
    print("Upload failed:", e)
```



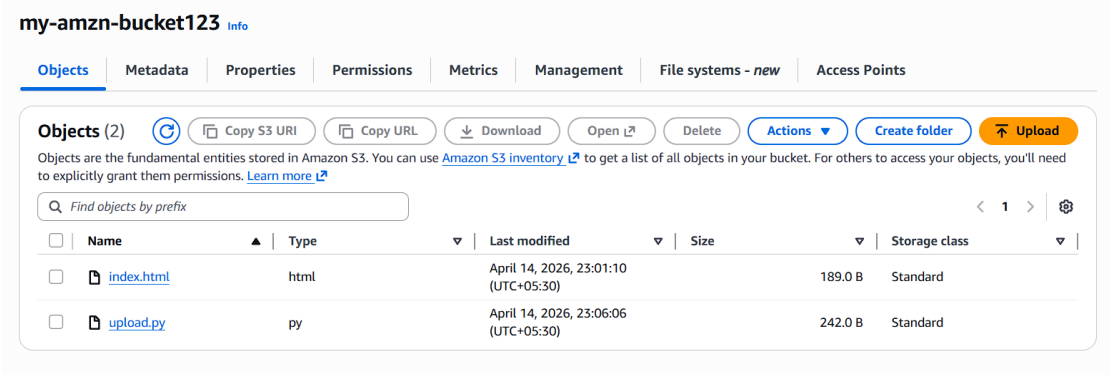
Step 3: Run Script

python upload.py

Output:

File uploaded successfully

The file will now appear inside the **S3 bucket**.



Result:-

The S3 bucket was successfully created, static website hosting was enabled, and the website was deployed and accessed through the S3 endpoint. Files were also uploaded programmatically using a Python script with Boto3.

Conclusion:-

This experiment demonstrates how Amazon S3 can be used as a lightweight web hosting platform. Using Python and Boto3, developers can automate file uploads and integrate S3 into applications.

Prof. S. D. Cheke
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